Chapter 1 – Why Normalization Exists

Chapter Overview

By the end of this chapter, you will understand:

- What is data?
- What is a database?
- Why databases become messy
- Why duplicate data is dangerous
- What is redundancy?
- What are anomalies?
- Why normalization was invented
- Real-world examples
- Introduction to Normal Forms

---

1. What is Data?

Before learning databases, understand **data**.

Data is simply a collection of facts.

Example:

```
Akshat
22
Delhi
9876543210
```

These are individual pieces of data.

Real-life examples of data

Student Data

```
Name
Age
Roll Number
Phone Number
Marks
```

Hospital Data

```
Patient Name
Doctor
Disease
Medicine
```

Amazon Data

```
Customer
Product
Price
Address
Order Date
```

Every company in the world stores data.

---

2. What is Information?

Data by itself has little meaning.

When data is processed and organized, it becomes **information**.

Example

Data

```
Marks = 92
```

Information

```
Akshat scored 92 marks in Database Management System.
```

Information gives meaning.

---

3. What is a Database?

A **Database** is an organized collection of related data.

Imagine a huge cupboard.

Instead of clothes, it stores information.

Example

A college stores

- Students
- Teachers
- Courses
- Attendance
- Fees
- Results

All this is stored in one database.

Example

Without Database

```
Student 1 information in notebook

Student 2 information in another notebook

Student 3 information in Excel

Teacher information somewhere else
```

Everything becomes messy.

With Database

Everything is stored in one organized system.

---

4. What is DBMS?

DBMS stands for

Database Management System

It is software that helps us create and manage databases.

Examples

- MySQL
- PostgreSQL
- Oracle
- SQL Server
- SQLite

Think of it like this.

Database = Books

DBMS = Librarian

The books are stored safely.

The librarian knows

- where each book is
- who borrowed it
- when it should return

Similarly,

MySQL manages your data.

---

5. Why Do Companies Use Databases?

Imagine Amazon.

Millions of customers.

Millions of products.

Millions of orders.

Can this be managed using Excel?

No.

A database helps

- Store data
- Retrieve data quickly
- Update data
- Delete data
- Maintain security
- Prevent duplication

---

6. What is a Table?

A database contains tables.

Example

Student Table

StudentID	Name	Age
101	Akshat	22
102	Rahul	20

Each row is called a **Record (Tuple)**.

Each column is called an **Attribute (Field)**.

Important Terminology

Term	Meaning
Table	Collection of rows and columns
Row	One complete record
Column	One property
Record	Row
Field	Column
Attribute	Column
Tuple	Row

These words are often used interchangeably.

7. The Biggest Problem in Databases

Suppose we make this table.

StudentID	Name	Course	Teacher
101	Akshat	SQL	Ramesh
101	Akshat	Python	Suresh
101	Akshat	Java	Amit

Notice something.

How many times is

Akshat repeated?

Three times.

StudentID?

Three times.

Is it wrong?

No.

But is it efficient?

No.

This problem is called

Redundancy

---

8. What is Redundancy?

Redundancy means

Storing the same information multiple times.

Example

StudentID	Name
101	Akshat
101	Akshat
101	Akshat

Same information repeated.

Real-life Example

Imagine your school asks you to write your address

100 times.

```
Delhi

Delhi

Delhi

Delhi

Delhi
```

Waste of time.

Waste of paper.

More chances of mistakes.

Same thing happens in databases.

Why is Redundancy Bad?

Because it creates

- More storage usage
- Slow updates
- Inconsistency
- Difficult maintenance

Example

Suppose

Akshat changes his name to

Akshat Saini.

Database

StudentID	Name
101	Akshat
101	Akshat
101	Akshat

You must update

ALL THREE ROWS.

If you update only two

StudentID	Name
101	Akshat Saini
101	Akshat Saini

StudentID	Name
101	Akshat

Now database contains

Two different names.

Which one is correct?

Nobody knows.

This is called

Data Inconsistency.

---

9. What is Data Inconsistency?

When the same information exists in multiple places but the values are different.

Example

StudentID	Name
101	Akshat
101	Akshat Saini

This should never happen.

---

10. Database Anomalies

Because of redundancy,

three major problems occur.

They are called

Database Anomalies.

---

1. Update Anomaly

Suppose

Teacher changes

Ramesh

↓

Ramesh Kumar

Table

Student	Subject	Teacher
Akshat	SQL	Ramesh

Student	Subject	Teacher
Rahul	SQL	Ramesh
Mohit	SQL	Ramesh

Need to update

Three rows.

If one row is missed

Database becomes inconsistent.

This is

Update Anomaly.

---

2. Insert Anomaly

Suppose

A new course starts.

Artificial Intelligence

Teacher

Neha

But

No student has enrolled yet.

Current table

Student	Subject	Teacher

Can we insert

Artificial Intelligence?

No.

Because Student column cannot be left empty.

So we cannot store information.

This is

Insert Anomaly.

---

3. Delete Anomaly

Suppose

Student	Subject	Teacher
Akshat	SQL	Ramesh

Akshat leaves college.

Delete this row.

Now

We also lost

Teacher = Ramesh teaches SQL.

Information lost.

This is

Delete Anomaly.

Summary of Anomalies

Anomaly	Meaning
Update	Must update many rows
Insert	Cannot insert some data independently
Delete	Deleting one row accidentally removes other important information

---

11. Why Was Normalization Invented?

Scientists noticed

Big tables create

- Redundancy
- Inconsistency
- Storage waste
- Difficult maintenance

So they created a technique called

Normalization

Definition

Normalization is the process of organizing data into multiple related tables to reduce redundancy and eliminate anomalies.

Simple definition:

> **Normalization means breaking a large, repetitive table into smaller, well-structured tables without losing information.**

---

12. Real-Life Analogy

Imagine your wardrobe.

Without organization:

```
Shoes
Books
Shirts
Laptop
Food
Files
```

Everything mixed together.

Hard to find anything.

Now organize it:

Shelf 1 → Clothes

Shelf 2 → Shoes

Shelf 3 → Books

Shelf 4 → Files

Everything becomes easier.

Normalization does the same thing for data.

---

13. Before Normalization

Student	Course	Teacher
Akshat	SQL	Ramesh
Akshat	Python	Suresh
Rahul	SQL	Ramesh

Notice

Teacher names repeat.

Student names repeat.

After Normalization

Student

StudentID	Name
101	Akshat
102	Rahul

Course

CourseID	Course

CourseID	Course
C1	SQL
C2	Python

Teacher

TeacherID	Name
T1	Ramesh
T2	Suresh

Enrollment

StudentID	CourseID
101	C1
101	C2
102	C1

Teaching

CourseID	TeacherID
C1	T1
C2	T2

No unnecessary repetition.

---

14. Benefits of Normalization

- Reduces duplicate data.
- Saves storage space.
- Prevents inconsistencies.
- Easier to update.
- Easier to maintain.
- Improves data integrity.
- Makes database design scalable.
- Reduces anomalies.

---

15. Levels of Normalization

Normalization is performed in stages called **Normal Forms**.

Normal Form	Purpose
1NF	Remove repeating groups and ensure atomic values
2NF	Remove partial dependencies
3NF	Remove transitive dependencies
BCNF	Every determinant must be a candidate key

Normal Form	Purpose
4NF	Remove multi-valued dependencies
5NF	Remove join dependencies

You don't jump directly to 3NF or BCNF. You move step by step:
1NF → 2NF → 3NF → BCNF...

Chapter 1 Recap

- **Data** = Raw facts.
- **Information** = Processed data with meaning.
- **Database** = Organized collection of related data.
- **DBMS** = Software to manage databases (e.g., MySQL).
- **Redundancy** = Unnecessary repetition of data.
- **Data inconsistency** = Same fact stored with different values.
- **Anomalies** = Problems caused by poor database design:
  - Update anomaly
  - Insert anomaly
  - Delete anomaly
- **Normalization** = Organizing data into well-structured tables to reduce redundancy and eliminate anomalies.

Practice Questions

1. What is the difference between data and information?
2. What is a database?
3. What is a DBMS? Give four examples.
4. What is redundancy? Explain with an example.
5. What is data inconsistency?
6. Explain update anomaly with a real-life example.
7. What is insert anomaly?
8. What is delete anomaly?
9. Define normalization.
10. Why is normalization important in relational databases?

Memory Tricks

- **Data** = Raw facts.
- **Information** = Meaningful data.
- **Database** = Organized storage.
- **Redundancy** = Repeated data.
- **Normalization** = Organize data to reduce repetition and anomalies.
- **Three anomalies** = **Update, Insert, Delete (UID)**.